

Tushar Jayanti

+91-9654907479 | tj@tusharjayanti.io | tusharjayanti.io | [GitHub](#) | [LinkedIn](#)

SUMMARY

Senior Backend Engineer (7+ years across fintech, legal tech, and e-commerce), designing and building production-grade Agentic AI using multi-agent orchestration, inline hallucination detection, and eval-gated CI. Currently developing a stealth venture focused on high-accuracy AI content detection.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C#, JavaScript, TypeScript

Backend: FastAPI, Flask, Django, Spring Boot, gRPC, REST, GraphQL, Kafka, Microservices, Event-Driven Architecture

Databases: PostgreSQL, MSSQL, MongoDB, Redis, Elasticsearch, DynamoDB

Cloud/DevOps: AWS, Docker, Kubernetes, Terraform, GitLab CI, Jenkins, Datadog

Data & Orchestration: Apache Airflow, Batch Processing, Real-time Event Systems

AI/LLM: Agentic AI, Multi-Agent Systems, RAG (hybrid pgvector + BM25, reranking), Vector Embeddings, LLM API integration, LLMops & Observability, Eval-Gated CI, Hallucination-Detection Eval Loops, Prompt Engineering, Prompt-Injection Defense, Pydantic, Claude Code, GitHub Copilot

EXPERIENCE

Stealth AI Content Detection Platform

Jan 2026 – Present

Engineering

Stealth Venture

- Building an **AI content-detection platform** that takes a piece of content and returns whether it is **AI-generated**, and which **model family** produced it, with **calibrated confidence** and audit-ready reporting, delivered as an **API** on **cloud-first** infrastructure.
- Targeting reliable detection that holds up under **humanizers and adversarial rewrites**, where the naive perplexity-based approaches most tools lean on break down, the core research and engineering challenge of the product.
- Designing for **model-family attribution** across a growing set of frontier models, text first with other modalities on the roadmap.
- Architecting the platform around **evaluation and observability** from day one, LLM-as-a-Judge scoring, attribution correctness, and end-to-end tracing, so detection quality is measured, not assumed.

DISCO

Nov 2023 – Sept 2025

Senior Software Engineer

Platform / Identity Systems

- Led migration of mission-critical identity platform from .NET + RavenDB to **Kotlin + Spring Boot + PostgreSQL**, improving scalability and long-term maintainability.
- Built and owned **multi-tenant, permissions-aware identity and authorization services** with audit logging, supporting enterprise-grade access control across thousands of customers.
- Designed and owned high-performance backend APIs and service-to-service communication using **Python (FastAPI/Flask) and gRPC with Protocol Buffers**.
- Migrated Python authorization service from REST to **gRPC with Protocol Buffers**, enabling strongly typed contracts and high-performance inter-service communication.
- Reduced **p99 latency from 4.2s to 1s (75%+ improvement)**, improving throughput and efficiency under production load.
- Built custom **LINQ-style C# → PostgreSQL query framework**, improving data access safety, consistency, and developer productivity.
- Improved **CI/CD speed by 67%** and reduced Docker image sizes by migrating **Python services from pipenv to uv**.
- Provisioned and managed infrastructure using **Terraform + AWS**, improving deployment consistency and system reliability.
- Mentored engineers and drove architectural best practices across platform services.

PurpleToko

Oct 2022 – Oct 2023

Founding Engineer

Distributed E-commerce Platform

- Built and iterated backend APIs and systems on AWS for search, order processing, and payments in a fast-paced startup environment.
- Implemented **Elasticsearch-based search** and **SQS-driven asynchronous pipelines** for resilient distributed processing.
- Designed stateless services enabling horizontal scaling and fault isolation in a 0→1 growth environment.
- Translated ambiguous product requirements into scalable backend systems and production-ready services.

Transcend Street Solutions

Sept 2020 – Sept 2022

Software Engineer

Financial Systems

- Engineered and launched the **Reserve Release** feature end-to-end, preventing incorrect financial orders across **10k+ daily transactions**.
- Optimized SQL workloads, reducing latency **8x (6s to 700ms)** through indexing and query optimization.
- Built and orchestrated batch data pipelines using **Apache Airflow**, enabling reliable and observable workflow execution.
- Migrated monolith modules to **Spring Boot microservices** and built REST APIs for high-throughput workflows.

Baanyan Software Services (USAA)

Full Stack Java Developer

Aug 2018 – Jul 2020

Financial Systems

- Developed Spring Boot services integrated with **Kafka-based event-driven architecture** for high-throughput financial processing.
- Created multiple plugins using Python to extend functionality for third-party tools.
- Built monitoring dashboards using **Prometheus and Grafana**, improving observability and system reliability.

PROJECTS

tusharjayanti.io

github.com/tusharjayanti/tusharjayanti.io

- Built a production **LLMOps** portfolio with an **agentic RAG** chat assistant on **Vercel Edge Functions** and **React 19/TypeScript**, using **Claude Sonnet** for generation, **Claude Haiku** for listwise reranking, **Voyage embeddings**, and hybrid **pgvector + BM25** retrieval over Supabase.
- Engineered end-to-end **observability with Langfuse** tracing, multi-layer **prompt-injection defense**, prompt caching, and an **eval-gated CI pipeline** (90-query eval suite, 213 tests) gating every deploy against a live production baseline.

vox-agent

github.com/tusharjayanti/vox-agent

- Built a production-grade **agentic customer support AI** using **FastAPI, asyncpg, and pgvector**, with an inline **hallucination-detection eval loop** using Claude Sonnet as generator and Claude Haiku as LLM judge.
- Designed end-to-end backend architecture: async API layer, vector similarity retrieval, and structured LLM orchestration with tool use.

shortlist

github.com/tusharjayanti/shortlist

- Built a **multi-agent** job-search copilot in **Python 3.12** orchestrating **six AI agents** (scoring, resume tailoring, cover-letter and outreach generation) with **Pydantic** typed schemas and **LLM provider-agnostic** support across Anthropic, OpenAI, and Gemini.
- Grounded every output in a structured career corpus to prevent hallucination, with **human-in-the-loop** review, **PostgreSQL**-backed resumable state, and audit logging on every LLM call.

Intelligent Chatbot

github.com/tusharjayanti/TensorflowChatbot

- Built a **seq2seq chatbot** using **Python and TensorFlow**, trained on a movie dialogue dataset to generate contextually relevant responses.

Optical Character Recognition

github.com/tusharjayanti/OpticalCharacterRecognition

- Developed an **OCR system** using **Linear and Logistic Regression in Python and TensorFlow** to improve accuracy on small fonts and noisy image data for CAPTCHA recognition.

EDUCATION

Master of Science in Computer Science — New Jersey Institute of Technology

Bachelor of Technology in Electrical Engineering — Guru Gobind Singh Indraprastha University